INFOSYS SPRINGBOARD

VIRTUAL INTERNSHIP 7.0

SMART URBAN MOBILITY & TRAFFIC INTELLIGENCE

PROJECT DOCUMENTATION

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Batch	3 (August)
Tool	Power BI
Project	Smart Urban Mobility & Traffic Intelligence
Year	2026

1. Project Objectives

The main objective of the **Smart Urban Mobility & Traffic Intelligence** project is to analyze urban mobility and traffic data using Power BI and convert the data into meaningful and easy-to-understand insights. The project aims to understand traffic patterns, transportation demand, route performance, mobility behavior, geographic patterns, and the factors that influence urban transportation.

The project also focuses on developing an interactive dashboard that helps users explore mobility data through visualizations, KPIs, filters, and analytical insights. The overall goal is to support better understanding of urban transportation patterns and help in data-driven mobility planning and decision-making.

2. Project Description

Smart Urban Mobility & Traffic Intelligence is a Power BI-based data analytics project designed to analyze and visualize urban transportation and traffic information. The project uses mobility-related data such as passenger demand, traffic volume, transportation modes, routes, travel time, speed, weather conditions, temperature, incidents, and geographic locations.

The data is processed, cleaned, organized, and modeled in Power BI before being presented through interactive dashboards. Different charts, KPI cards, maps, slicers, and analytical features are used to identify patterns and relationships within the data.

The project provides a consolidated view of urban mobility and traffic conditions, making complex transportation data easier to understand and supporting better analysis of demand, routes, transportation modes, weather effects, and mobility trends.

3. Tools Used

Tool / Technology	Purpose
Microsoft Power BI Desktop	Data modeling, DAX, dashboard development, visualization and interactivity.
Power Query	Data cleaning, transformation, merging and preparation of supporting tables.
DAX	Measures and calculations for traffic, trips, passengers, speed, travel time and reliability.
Icon Map Pro	Geospatial heatmap, flow-map lines and geographic map visualization.
CSV / Excel dataset	Source urban mobility data used for the project.

4. Milestone 1 – Data Integration & Data Modeling

4.1 Milestone 1 Requirements

The requirements given for Milestone 1 were:

- Acquire and integrate urban mobility, transit, weather, and demographic datasets.
- Perform ETL operations and data cleansing using Power Query.
- Design and implement a star-schema data model.
- Create relationships between the required tables.
- Develop DAX functions and calculations for mobility analysis.
- Create KPI cards for important mobility metrics.
- Develop bar/column charts for comparative analysis.
- Create pie/donut charts for distribution analysis.
- Create line charts for trend analysis.
- Load, validate, and analyze the prepared data in Power BI.

4.2 Milestone 1 Implementation

I started by importing the urban mobility dataset into Power BI and reviewing the available fields. The dataset contained information related to routes, locations, passenger count, traffic volume, vehicles, average speed, travel time, traffic incidents, weather and other mobility-related factors.

I used Power Query to perform the required data preparation and cleansing. The data types, duplicate records, missing values and formatting consistency were checked and corrected where required.

After cleaning the data, I created a structured data model using fact_traffic as the main fact table and supporting dimension tables such as dim_date, dim_route and dim_weather. Relationships were created between the tables to support filtering and analysis.

I then created DAX measures such as Total Traffic Volume, Total Traffic Incidents, Total Passenger Count, Average Travel Time and Average Speed. These measures were used throughout the dashboard to provide reusable and consistent calculations.

Finally, I developed the Milestone 1 dashboard using KPI cards, charts, slicers and other Power BI visuals. The dashboard was validated by checking the calculated results and ensuring that the relationships and filters worked correctly.

4.3 Milestone 1 Dashboard Explanation

The Milestone 1 dashboard provides an overall view of urban mobility and traffic conditions using the cleaned and structured dataset. The main purpose of this dashboard was to convert the raw mobility data into an interactive report where important traffic information could be understood quickly.

The dashboard begins with **KPI cards** that display the major mobility indicators. The **Total Traffic Volume** card represents the overall number of vehicles recorded in the dataset. The **Total Traffic Incidents** card provides the total number of reported traffic incidents. The **Total Passenger Count** card represents the overall passenger demand recorded across the available routes and dates. The **Average Travel Time** card provides the average time taken for travel across the recorded trips. These KPI cards provide a quick summary before moving into detailed visual analysis.

The **Average Speed by RouteID** visual is used to compare the average travel speed across different routes. This helps in understanding how traffic movement varies from one route to another. Instead of using the sum of speed values, the average speed measure was used because speed is a continuous measurement, and an average provides a more meaningful comparison between routes.

The **Traffic Volume by RouteID** visual provides a route-level comparison of traffic volume. It helps identify routes with relatively higher or lower traffic activity. This allows users to understand the distribution of traffic across the different routes included in the dataset.

The **Traffic Level Distribution** visual represents the proportion of different traffic conditions in the dataset. It provides a quick visual understanding of how frequently different traffic levels occur. This makes it easier to identify the overall traffic condition distribution without examining individual records.

The **date-based line chart** shows how mobility or traffic values change over time. By observing the trend across different dates, users can identify periods where traffic activity increases or decreases. This provides a basic temporal understanding of urban mobility patterns.

Interactive **slicers** were included to make the dashboard dynamic. The **Date slicer** allows users to focus on a specific date or time period. The **Weather slicer** allows analysis under selected weather conditions, while the **RouteID slicer** allows users to focus on individual routes.

The dashboard was designed with a consistent layout, clear titles, KPI cards and properly spaced visuals. The combination of summary indicators, comparative charts, distribution analysis, trend analysis and slicers make the dashboard easier to explore.

Overall, the Milestone 1 dashboard established the basic analytical foundation of the project. It allowed the user to move from raw mobility records toward a structured view of **traffic volume, passenger demand, incidents, travel time, speed and traffic conditions**. The data model and measures created here were reused and extended in the later milestones.

5. Milestone 1 – Output Screenshots

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF
1	RouteID	Source Loc	Destinatio	Date	Time	Day	Month	Quarter	Passenger	Traffic Vol	Cars	Trucks	Bikes	Buses	Average Sp	Travel Time	Traffic Lev	Weather	Temperatu	Humidity	Rainfall	Accident R	Traffic Inci	Signal Stat	Population	Population	Road Conc	Public Tran	AQI	Peak Hour	Time Perio	Is Weekend
2	RT0001	LB Nagar	Koti	########	00:00	Tuesday	March	Q1	1216	366	155	38	121	52	32	36.6	Low	Clear	20	40	0.7	No	0	Malfunctio	738508	9515	Poor	959	96	No	Night	No
3	RT0002	Gachibowl	Hitech City	########	21:00	Wednesda	July	Q3	592	505	246	53	192	14	33	14.2	Medium	Clear	31	89	1	No	0	Working	631741	20571	Poor	355	136	No	Night	No
4	RT0003	Dilsukhna	Abids	########	03:30	Sunday	July	Q3	1201	266	128	18	73	47	32	48.5	Low	Clear	32	52	1.4	No	1	Malfunctio	538162	14641	Poor	529	211	No	Night	Yes
5	RT0004	Kondapur	Gachibowl	########	17:15	Tuesday	February	Q1	1140	661	453	41	133	34	35	41	Medium	Cloudy	30	88	2.3	Yes	1	Working	480776	16145	Poor	648	56	Yes	Evening	No
6	RT0005	Uppal	Hitech City	########	19:15	Sunday	August	Q3	605	494	282	68	123	21	32	34.3	Medium	Foggy	33	72	1.2	No	1	Working	684277	19171	Good	286	233	Yes	Evening	Yes
7	RT0006	Mehdipatn	Jubilee Hill	########	08:15	Tuesday	April	Q2	1372	679	385	18	247	29	31	21.3	Medium	Foggy	20	78	2.2	No	4	Malfunctio	429786	24003	Average	796	68	Yes	Morning	No
8	RT0007	Hyderabad	Secundera	########	23:45	Friday	April	Q2	987	416	164	74	141	37	39	47.9	Medium	Foggy	26	44	1.1	No	4	Working	150599	22626	Good	763	165	No	Night	No
9	RT0008	Miyapur	Ameerpet	########	19:30	Friday	Septembe	Q3	654	499	109	40	340	10	17	54	Medium	Foggy	24	43	2	No	1	Under Mair	427932	20290	Poor	966	94	Yes	Evening	No
10	RT0009	Mehdipatn	Jubilee Hill	########	07:30	Sunday	May	Q2	991	634	221	66	314	33	37	12.9	Medium	Clear	38	70	0.7	No	0	Working	892225	23679	Good	220	98	No	Early Morn	Yes
11	RT0010	Kondapur	Gachibowl	########	13:00	Thursday	August	Q3	927	512	151	45	298	18	53	52.1	Medium	Foggy	38	65	0.7	No	3	Malfunctio	349659	6090	Average	298	208	No	Afternoon	No
12	RT0011	Uppal	Hitech City	########	03:45	Friday	October	Q4	878	190	57	22	81	30	37	18.6	Low	Foggy	34	43	1.3	No	1	Under Mair	229046	17520	Poor	300	52	No	Night	No
13	RT0012	Miyapur	Ameerpet	########	01:00	Saturday	January	Q1	1155	510	115	62	298	35	58	12.6	Medium	Partly Clou	20	59	0.8	No	3	Working	449105	16861	Average	1096	79	No	Night	Yes
14	RT0013	Dilsukhna	Abids	########	14:00	Thursday	May	Q2	485	198	61	50	79	8	46	51.4	Low	Foggy	25	38	2.9	No	1	Working	221849	22498	Average	239	212	No	Afternoon	No
15	RT0014	Dilsukhna	Abids	########	20:15	Sunday	March	Q1	1278	521	384	15	91	31	31	24.2	Medium	Rainy	26	77	17.9	No	3	Malfunctio	287235	24151	Good	714	157	Yes	Evening	Yes
16	RT0015	Dilsukhna	Abids	########	08:45	Tuesday	January	Q1	1055	516	368	22	87	39	23	16	Medium	Rainy	22	50	9.2	No	4	Under Mair	887715	12912	Poor	1183	42	Yes	Morning	No
17	RT0016	Gachibowl	Hitech City	########	09:00	Thursday	July	Q3	1827	395	215	24	104	52	23	22.7	Low	Cloudy	30	48	2.1	No	4	Under Mair	662262	11228	Average	204	63	Yes	Morning	No
18	RT0017	Hyderabad	Secundera	########	21:00	Tuesday	July	Q3	780	425	200	26	184	15	52	29.2	Medium	Clear	23	39	2.8	No	1	Under Mair	722092	4180	Good	856	189	No	Night	No
19	RT0018	Hyderabad	Secundera	########	09:15	Tuesday	October	Q4	720	390	237	56	70	27	18	25.9	Low	Foggy	33	74	2.2	No	1	Working	320395	8801	Poor	944	46	Yes	Morning	No
20	RT0019	Miyapur	Ameerpet	########	18:45	Friday	August	Q3	709	380	216	30	105	29	22	19	Low	Partly Clou	31	54	2.5	No	1	Malfunctio	174813	24627	Average	495	142	Yes	Evening	No
21	RT0020	Begumpet	Banjara Hil	########	09:30	Monday	February	Q1	1894	785	408	75	254	48	15	15.2	High	Rainy	25	72	24.1	No	0	Malfunctio	775550	17239	Average	807	226	Yes	Morning	No
22	RT0021	Mehdipatn	Jubilee Hill	########	03:00	Tuesday	December	Q4	1492	291	127	42	72	50	46	51.7	Low	Foggy	26	58	1.3	No	2	Malfunctio	803425	13286	Poor	355	224	No	Night	No
23	RT0022	Begumpet	Banjara Hil	########	03:45	Thursday	July	Q3	733	629	236	47	333	13	51	52.3	Medium	Cloudy	38	54	1.2	No	2	Under Mair	450850	9887	Poor	980	241	No	Night	No
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25	RT0024	Dilsukhna	Abids	########	03:45	Tuesday	June	Q2	1816	442	67	68	262	45	52	41.3	Medium	Partly Clou	32	50	0.4	No	0	Under Mair	595798	10170	Poor	460	245	No	Night	No
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27	RT0026	Miyapur	Ameerpet	########	07:45	Wednesda	July	Q3	1369	375	170	66	89	50	38	25.1	Low	Foggy	22	43	0.5	No	1	Malfunctio	890734	10010	Average	231	146	No	Early Morn	No
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29	RT0028	Hyderabad	Secundera	########	12:45	Saturday	October	Q4	1416	590	229	53	257	51	59	15.7	Medium	Partly Clou	37	36	2.7	No	5	Malfunctio	178418	5750	Good	977	74	No	Afternoon	Yes
30	RT0029	Mehdipatn	Jubilee Hill	########	07:30	Tuesday	February	Q1	1024	541	197	37	282	25	38	43.8	Medium	Partly Clou	28	88	0.2	Yes	1	Under Mair	516960	10347	Good	240	239	No	Early Morn	No
31	RT0030	Miyapur	Ameerpet	########	04:15	Wednesda	January	Q1	649	277	40	89	128	20	51	15.1	Low	Cloudy	34	79	0.8	No	4	Working	786745	6753	Poor	435	119	No	Night	No
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33	RT0032	Mehdipatn	Jubilee Hill	########	22:00	Thursday	August	Q3	1388	602	280	23	271	28	52	16.9	Medium	Cloudy	36	76	0.8	No	4	Under Mair	656983	18232	Average	992	227	No	Night	No
34	RT0033	Kukatpally	Madhapur	########	00:15	Saturday	August	Q3	1876	466	172	67	174	53	49	40.1	Medium	Rainy	20	66	21.3	No	1	Under Mair	522060	11465	Poor	797	111	No	Night	Yes
35	RT0034	Kondapur	Gachibowl	########	14:00	Tuesday	July	Q3	1746	371	127	20	173	51	47	44.1	Low	Partly Clou	35	63	2.4	No	1	Under Mair	574045	10972	Average	727	209	No	Afternoon	No
36	RT0035	Mehdipatn	Jubilee Hill	########	01:30	Thursday	November	Q4	894	558	199	55	282	22	37	15.4	Medium	Cloudy	30	42	2.2	No	5	Malfunctio	344143	9276	Average	543	229	No	Night	No
37	RT0036	Kukatpally	Madhapur	########	03:30	Thursday	December	Q4	823	370	129	47	166	28	30	41.9	Low	Cloudy	28	37	2.9	No	5	Working	282413	23901	Poor	1105	66	No	Night	No
38	RT0037	Uppal	Hitech City	########	02:45	Monday	July	Q3	470	460	255	53	144	8	33	47	Medium	Partly Clou	35	39	1.7	No	1	Working	306445	21443	Good	722	61	No	Night	No
39	RT0038	Kondapur	Gachibowl	########	02:15	Sunday	October	Q4	1197	591	224	67	276	24	39	35.6	Medium	Clear	23	83	0.6	No	5	Under Mair	235131	8146	Good	591	84	No	Night	Yes
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41	RT0040	LB Nagar	Koti	########	11:15	Tuesday	June	Q2	612	216	102	19	80	15	53	47.1	Low	Rainy	34	42	11.7	No	3	Working	435470	19399	Poor	1111	152	No	Morning	No
42	RT0041	Begumpet	Banjara Hil	########	05:45	Tuesday	January	Q1	893	286	158	13	96	19	60	10.9	Low	Rainy	38	37	19.1	No	4	Under Mair	833157	17490	Average	669	86	No	Early Morn	No
43	RT0042	Uppal	Hitech City	########	16:00	Tuesday	December	Q4	731	515	208	62	220	25	40	28.5	Medium	Partly Clou	29	77	2.8	No	4	Malfunctio	188479	17903	Average	280	120	No	Afternoon	No
44	RT0043	Kondapur	Gachibowl	########	03:45	Friday	March	Q1	850	426	30	79	286	31	46	26.3	Medium	Partly Clou	34	83	0.2	No	1	Working	452020	17356	Poor	1092	71	No	Night	No

Figure 1. Original dataset Dimensions: 500 Rows × 32 Columns

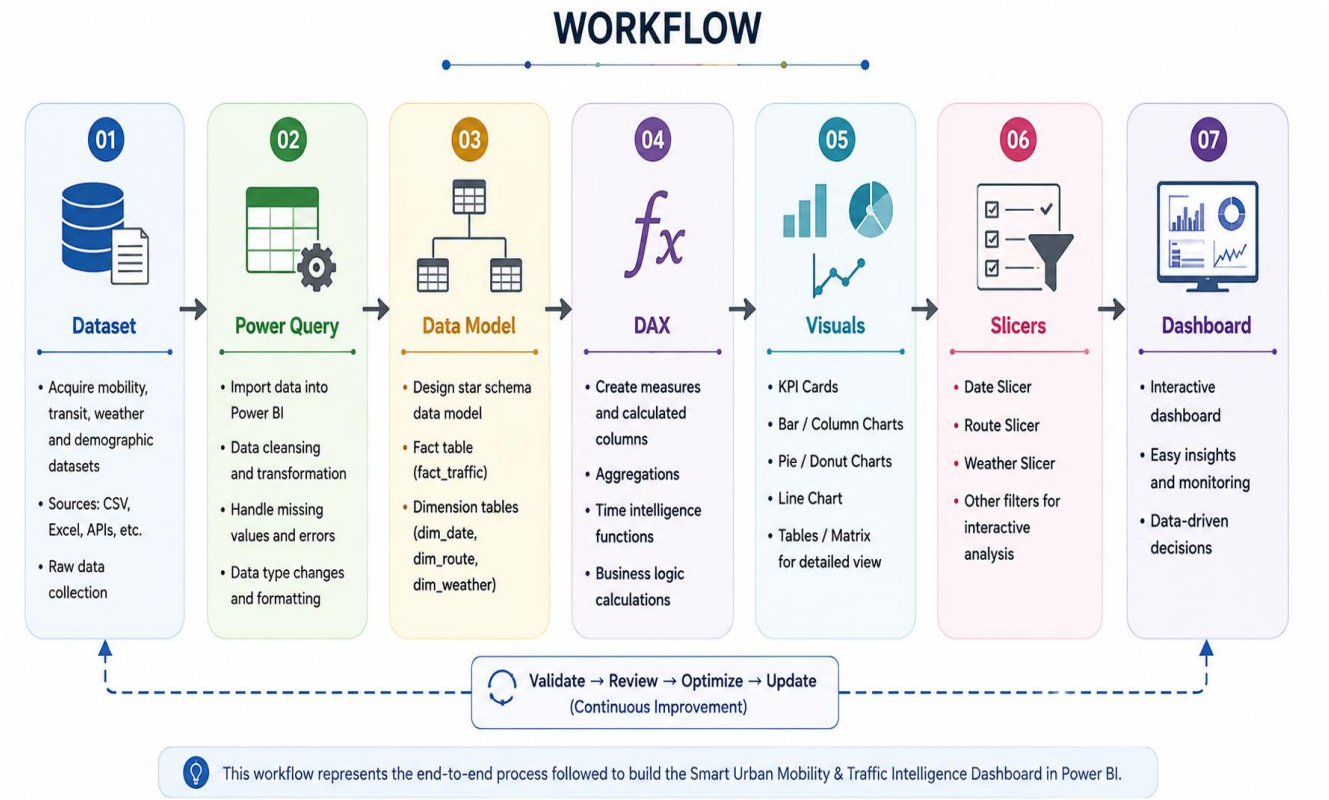


Figure 2. Milestone 1 workflow.

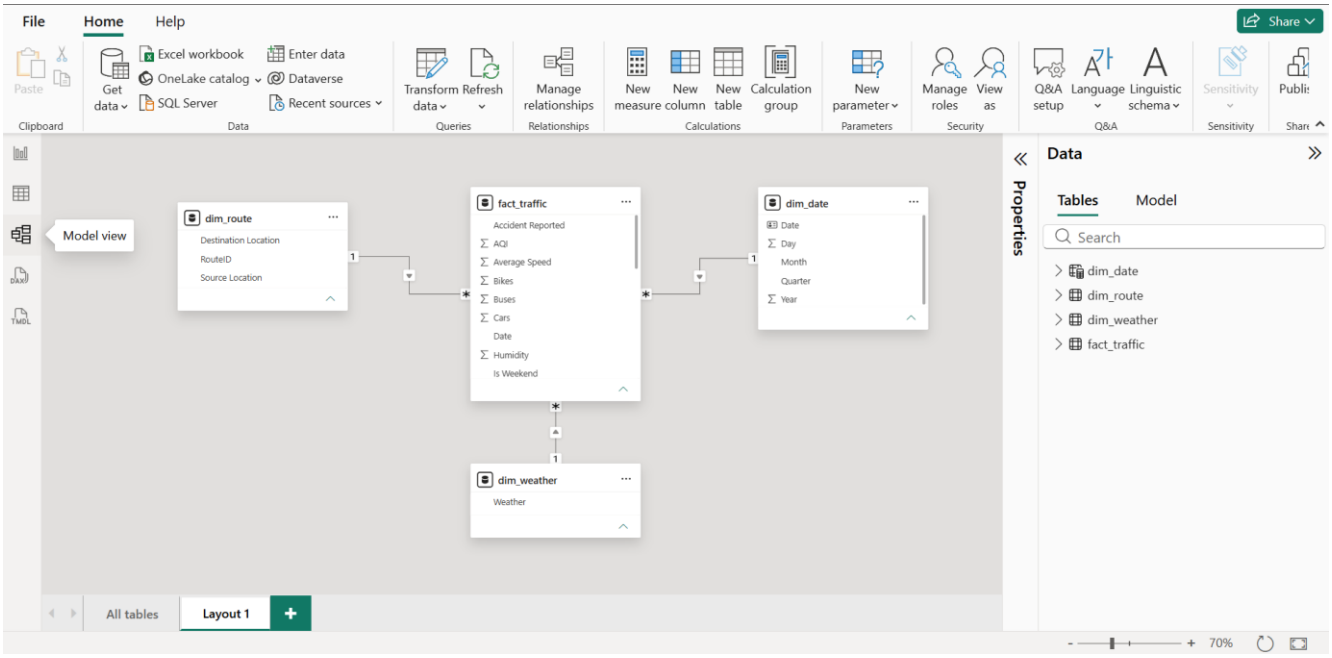


Figure 3. Table relationships.

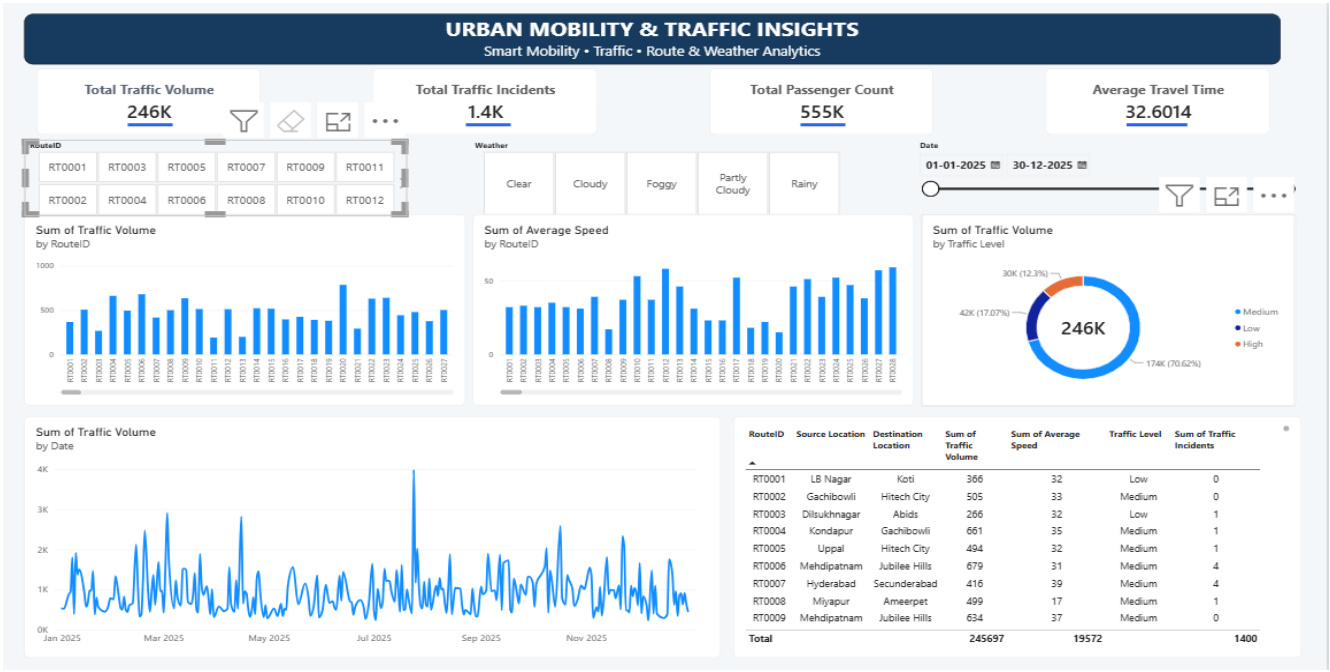


Figure 4. Final Milestone 1 dashboard

6. Milestone 2 – Geospatial & Transit Analytics

6.1 Milestone 2 Requirements

The requirements given for Milestone 2 were:

- Build mobility demand heatmaps and flow maps.
- Implement route performance and reliability dashboards.

- Develop geographic drill-down analysis.
- Integrate weather-related analysis.
- Apply temporal filters for mobility analysis.
- Perform geospatial and demand analysis.
- Analyze route performance.
- Add Date, Route, Location and Weather slicers.

6.2 Milestone 2 Implementation

In Milestone 2, I extended the foundation created in Milestone 1 by adding geographic and transit-focused analysis.

I created a dim_location table containing Location, City, Region, Latitude and Longitude information. I also prepared source, and destination coordinates for the routes so that geographic movement could be represented on the map.

A relationship was created between dim_location and dim_route to allow location-based filtering. I then developed geographic visuals to represent passenger demand and route connectivity across Hyderabad.

The dashboard was also extended with additional KPI measures for Total Traffic Incidents, Total Trips, Total Passenger Count and Traffic Reliability. Route performance and speed trends were analyzed using charts.

Weather and time-based analysis were added to understand how mobility conditions changed under different circumstances. Interactive slicers were included to make the dashboard easier to explore.

6.3 Milestone 2 Dashboard Explanation

The Milestone 2 dashboard extends the analysis from Milestone 1 by introducing **geographic, route, transit and weather-based analysis**. The main purpose of this dashboard was to understand not only what was happening in the mobility data, but also **where the activity was occurring and how different locations and routes were connected**.

The dashboard starts with four **KPI cards** that provide a quick summary of important mobility indicators. The **Total Traffic Incidents** card shows the overall number of recorded traffic incidents. The **Total Trips** card represents the number of mobility records or trips available in the dataset. The **Total Passenger Count** card provides an overview of passenger demand. The **Traffic Reliability %** card provides a summary measure of the reliability of the traffic system based on the available traffic information.

The **Average Speed Over Time** line chart shows how average travel speed changes across different dates. This visual helps users understand whether mobility performance remains relatively stable or changes during different periods.

The **Route Performance by Average Speed** bar chart compares the average speed of different routes. This allows users to examine route-level performance and identify differences in movement speed across the network. Data labels were added to make the values easier to read directly from the chart.

The **Mobility Demand Heatmap** provides a geographic representation of passenger demand. Location coordinates were prepared and connected to the mobility data so that demand could be displayed spatially. Areas with higher demand are represented by stronger heat concentrations, making it easier to identify locations where mobility activity is more concentrated.

The **Mobility Flow Map** focuses on route connectivity. Source and destination coordinates were prepared for the routes, allowing movement between locations to be represented visually. This helps users understand how different locations are connected through the mobility network.

The **Geographic Mobility Drill-down** visual provides another level of geographic analysis. A geographic hierarchy containing **Region, City and Location** was created so that users can move from a broader geographic level toward individual locations. This supports detailed analysis without requiring separate dashboards for each geographic level.

The **Weather Impact on Travel Time** visual compares average travel time under different weather conditions. This provides an easy way to examine whether travel time varies depending on the recorded weather category.

The dashboard also includes interactive **Date, Time, RouteID, Location and Weather slicers**. These filters allow users to narrow the analysis based on a particular period, route, location or weather condition. For example, a user can select a particular location and weather condition to examine the corresponding mobility patterns.

The geographic data preparation was an important part of this milestone. A separate dim_location table was created containing location, city, region, latitude and longitude information. Source and destination coordinates were also prepared for route mapping. These additions allowed the dashboard to move beyond traditional charts and represent mobility information geographically.

Overall, Milestone 2 transformed the initial traffic dashboard into a more detailed **geospatial and transit analytics dashboard**. It helped provide an understanding of **where passenger demand occurs, how routes connect locations, how routes perform, and how weather and time relate to mobility conditions**.

7. Milestone 2 – Output Screenshots



Figure 1. Milestone 2 workflow.

	Location	City	Region	Latitude	Longitude
1	LB Nagar	Hyderabad	Telangana	17.3457	78.5522
2	Gachibowli	Hyderabad	Telangana	17.4401	78.3489
3	Dilsukhnagar	Hyderabad	Telangana	17.3688	78.5247
4	Kondapur	Hyderabad	Telangana	17.458	78.367
5	Uppal	Hyderabad	Telangana	17.4058	78.5591
6	Mehdipatnam	Hyderabad	Telangana	17.3947	78.431
7	Hyderabad	Hyderabad	Telangana	17.385	78.4867
8	Miyapur	Hyderabad	Telangana	17.4969	78.357
9	Begumpet	Hyderabad	Telangana	17.4448	78.4667
10	Kukatpally	Hyderabad	Telangana	17.4849	78.4138
11	Koti	Hyderabad	Telangana	17.3849	78.4864
12	Hitech City	Hyderabad	Telangana	17.4435	78.3772
13	Abids	Hyderabad	Telangana	17.393	78.473
14	Jubilee Hills	Hyderabad	Telangana	17.4326	78.4071
15	Secunderabad	Secunderabad	Telangana	17.4399	78.4983
16	Ameerpet	Hyderabad	Telangana	17.4375	78.4483
17	Banjara Hills	Hyderabad	Telangana	17.4156	78.4347
18	Madhapur	Hyderabad	Telangana	17.4483	78.3915

Figure 2. Geographic location table (dim_location) with latitude and longitude.

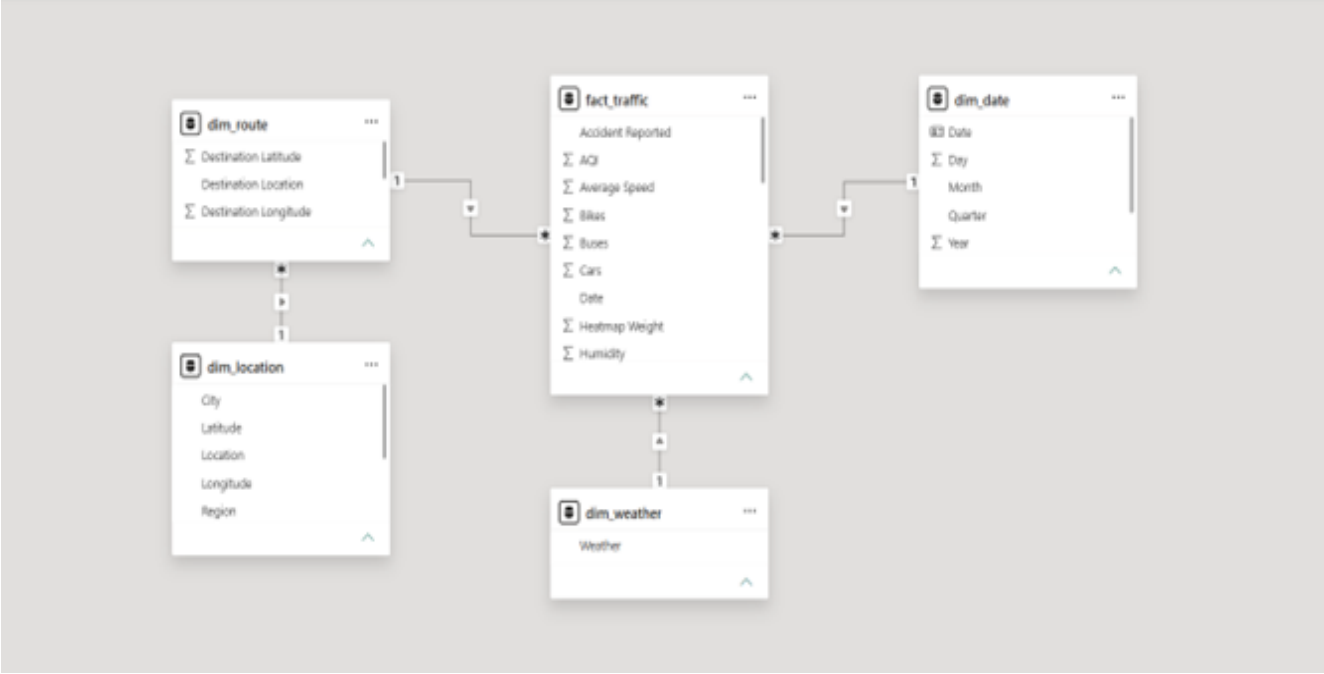


Figure 3. dim_location and its relationship with dim_route.

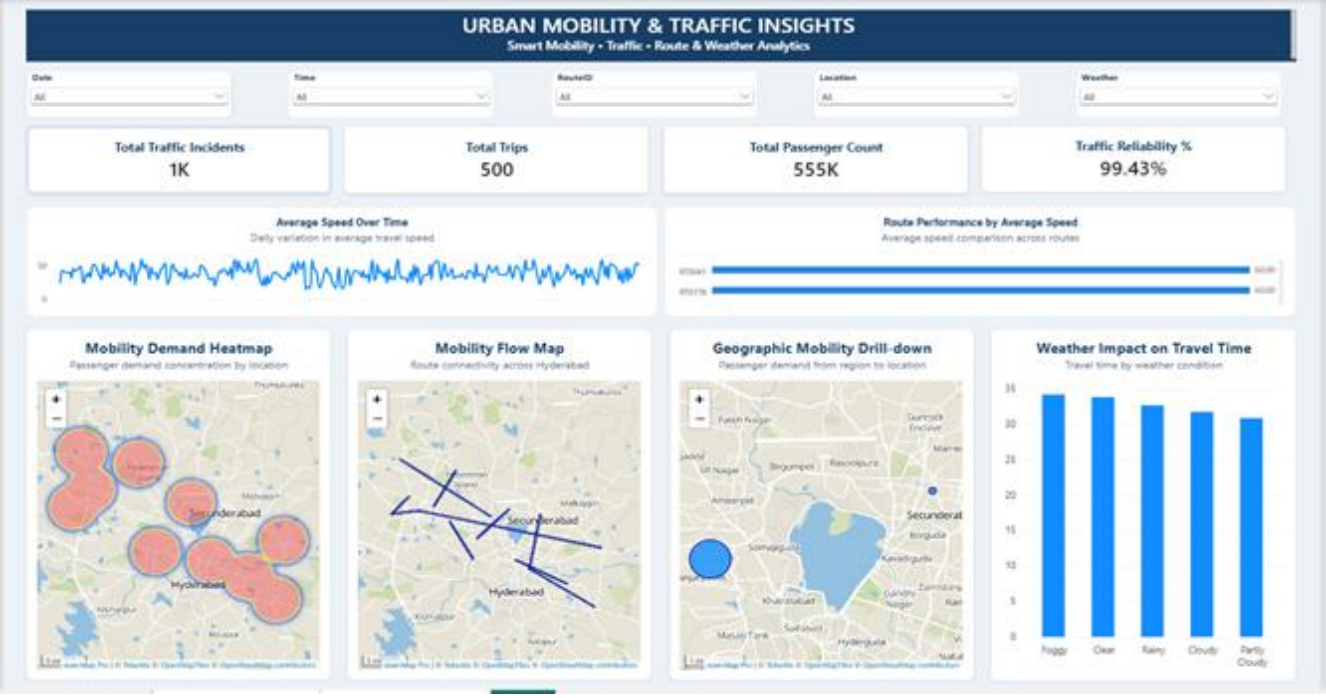


Figure 4. Final Milestone 2 dashboard.

8. Milestone 3 – Demand Intelligence & Mobility Analytics

8.1 Milestone 3 Requirements

The requirements given for Milestone 3 were:

- Build ride-hailing intelligence dashboards.
- Develop cycling and micromobility analytics.
- Implement modal substitution analysis.
- Add weather sensitivity analysis.
- Develop demand forecasting metrics.
- Analyze mobility demand across different time periods.
- Compare different mobility modes.
- Analyze the relationship between weather and demand.
- Develop demand trends and forecasting.

8.2 Milestone 3 Implementation

In Milestone 3, I extended the existing mobility dashboard to focus on demand intelligence and mobility analytics.

Since the available dataset did not contain specific ride-hailing fields such as service provider, fare or surge multiplier, I used the available mobility fields to perform demand-oriented analysis without introducing unsupported data.

I created DAX measures for Total Demand, Total Bike Volume, Total Car Volume, Total Bus Volume, Total Public Transport Usage, Total Truck Volume, Peak Hour Demand, Weekend Demand and Weekday Demand. Existing measures such as Average Speed and Average Travel Time were also reused.

The dashboard was developed with visuals for demand by time, demand by weather, bike volume by route, mobility mode comparison, temperature versus demand, and demand trends.

A forecasting feature was also added to the demand trend visual using Power BI's built-in forecasting capability. This provided an estimated continuation of the observed demand trend based on the available historical data.

8.3 Milestone 3 Dashboard Explanation

The Milestone 3 dashboard focuses on **demand intelligence and mobility analytics**. It builds on the data model and analysis developed in the previous milestones and introduces demand, mobility mode, cycling, weather sensitivity and forecasting analysis.

The dashboard begins with four **KPI cards** that summarize the main mobility demand indicators. The **Total Demand** card represents the total passenger demand recorded in the dataset. The **Total Bike Volume** card represents the total number of recorded bikes. The **Total Car Volume** card represents the total car volume, while the **Total Bus Volume** card represents the recorded bus volume. These KPIs provide a quick understanding of the overall demand and vehicle-mode activity.

The **Mobility Demand by Time** chart shows how passenger demand changes across different recorded times. This visual helps identify periods with relatively higher or lower mobility demand. It is particularly useful for understanding the temporal distribution of demand and supporting peak-period analysis.

The **Demand by Weather** chart compares passenger demand across different weather conditions. This provides a simple way to understand whether demand varies under different weather categories. The visual supports the weather sensitivity requirement by connecting mobility demand with environmental conditions.

The **Bike Volume by Route** chart focuses on cycling and micromobility analysis. It compares bike volume across different routes, allowing users to identify routes with higher or lower recorded cycling activity. This provides a route-level view of bike usage using the available dataset.

The **Mode Comparison** chart compares **cars, bikes and buses** across different routes. This visual supports modal analysis by showing how different transport modes contribute to mobility activity. Users can compare the relative volume of different modes and observe how the pattern changes from one route to another.

The **Temperature vs Demand** scatter chart examines the relationship between temperature and passenger demand. Temperature is placed on the horizontal axis and total demand on the vertical axis. Each point represents an observation from the dataset. The distribution of points helps users visually examine whether demand changes as temperature varies.

The **Demand Trend** line chart presents passenger demand over time. It allows users to observe changes in demand across the available dates. A **forecast** was also added to the visual using Power BI's forecasting functionality. This extends the analysis beyond historical demand and provides an estimated continuation of the observed trend.

Interactive **Date, RouteID, Weather, Time and Location slicers** were added to allow users to filter the dashboard according to their analysis requirements. These slicers make it possible to examine demand for specific dates, routes, weather conditions, times or locations.

The dashboard uses the existing mobility data model and DAX measures created for demand and transport modes. Measures such as **Total Demand, Total Bike Volume, Total Car Volume, Total Bus Volume, Total Public Transport Usage, Total Truck Volume, Peak Hour Demand, Weekend Demand and Weekday Demand** support the calculations used throughout the analysis.

The available dataset did not contain dedicated ride-hailing fields such as **service provider, average fare or surge multiplier**. Therefore, the dashboard does not introduce fabricated ride-hailing values. Instead, demand and mobility intelligence was developed using the actual passenger, vehicle, time, weather and route fields available in the dataset.

Overall, Milestone 3 expanded the project from geographic and traffic analysis into **demand intelligence, cycling and mobility-mode analysis, weather sensitivity and demand forecasting**. The dashboard provides a more

complete view of how mobility demand changes across **time, routes, transport modes, weather and temperature**, while the forecasting visual adds a forward-looking component to the analysis.

9. Milestone 3 – Output Screenshots

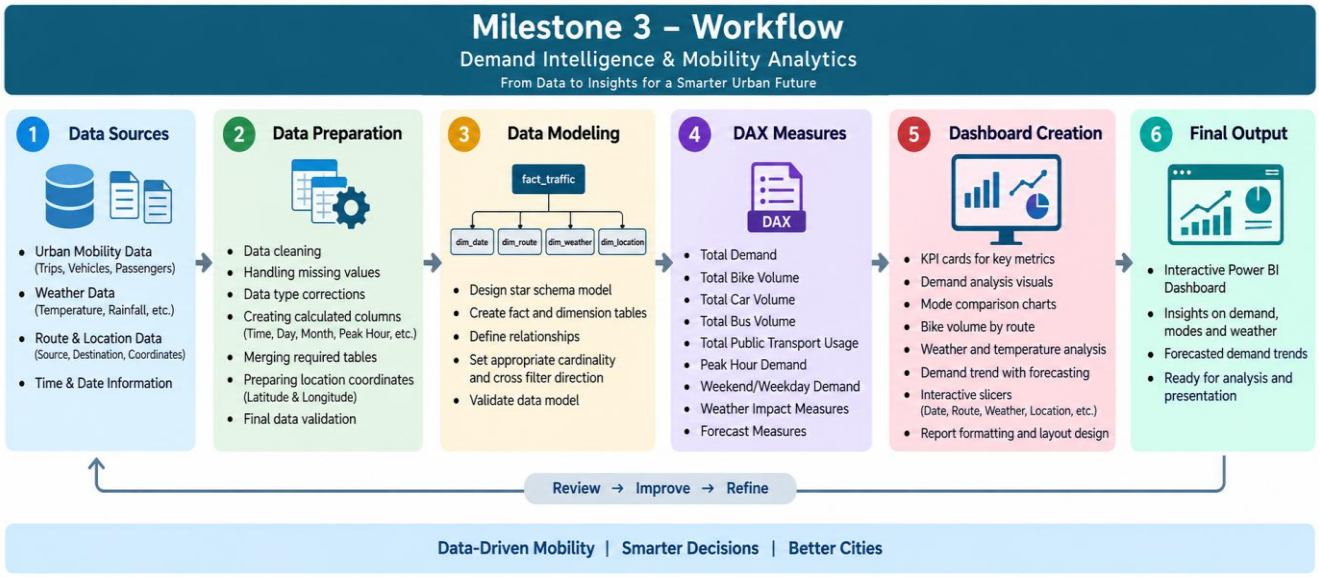


Figure 1. Milestone 3 workflow

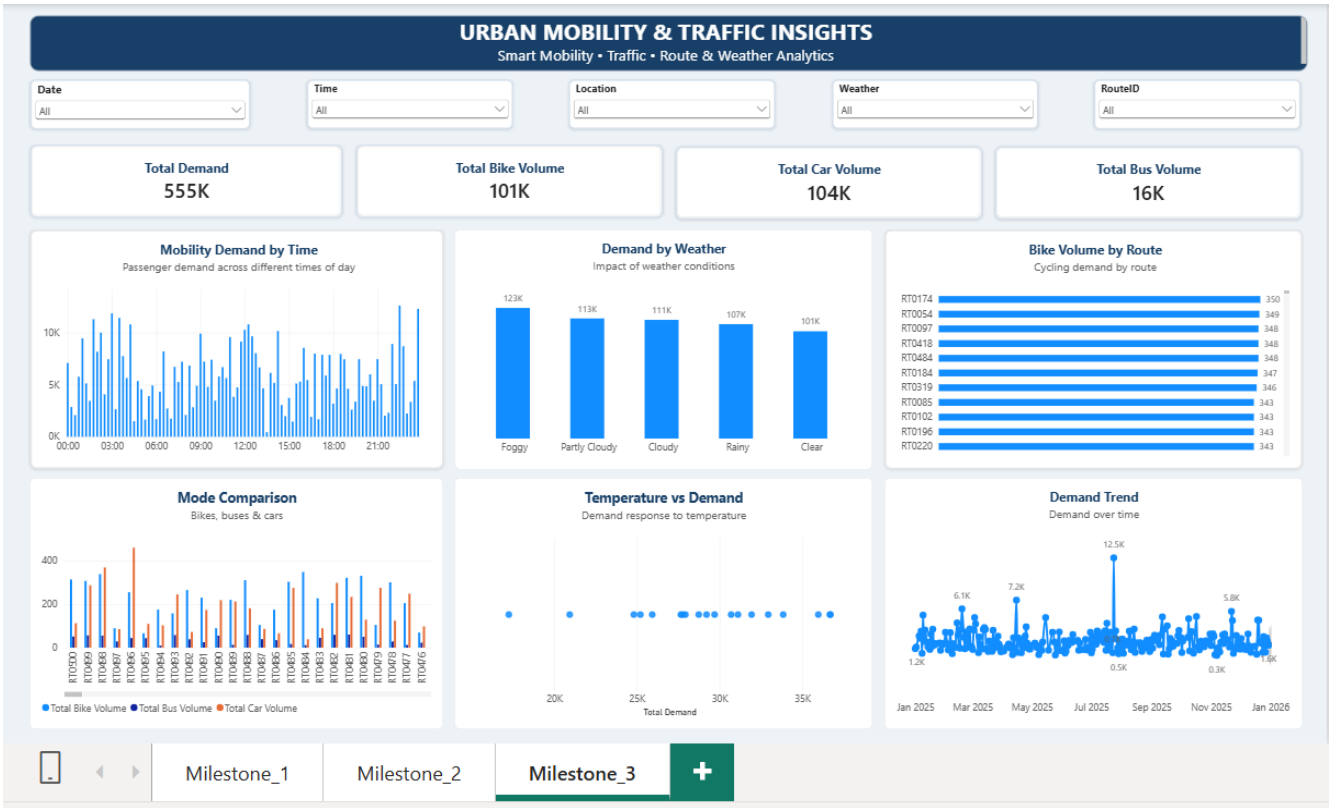


Figure 2. Final Milestone 3 dashboard.

9. Milestone 4 – Equity, Forecasting & Finalization

9.1 Milestone 4 Requirements

The requirements given for Milestone 4 were:

- Implement Mobility Access Index and equity dashboards.
- Build executive intelligence dashboards.
- Add forecasting, anomaly detection and reporting features.
- Optimize dashboard performance and complete deployment readiness.
- Develop demand and transport-mode intelligence.
- Analyze mobility access across different areas.
- Develop demand forecasting and anomaly detection.
- Prepare final project insights and conclusion.

9.2 Milestone 4 Implementation

In Milestone 4, I extended the existing mobility analytics dashboard by adding **executive intelligence, mobility equity, demand forecasting and final reporting**.

I created four new dashboard pages: **Executive Intelligence, Mobility Equity, Demand Forecast and Final Insights**. The existing data model, DAX measures and geographic relationships developed in the previous milestones were reused to maintain consistency across the project.

For the **Executive Intelligence** dashboard, I created additional measures for Total Trips, Traffic Reliability, Total Vehicle Volume and Mode Reliability. These measures were used to provide a high-level summary of passenger demand, trips, reliability and travel-time conditions.

For the **Mobility Equity** analysis, I created a project-defined **Mobility Access Index** using available demand, public transport usage and traffic reliability indicators. This index was used to compare mobility access across different locations.

The original requirement included **Average Fare** and **Mobility Access by Demographic Group**. However, the available dataset did not contain fare or demographic information. Therefore, I did not introduce unsupported values. **Public Transport Usage** was used as an available mobility-access indicator, while demographic-level analysis was not populated.

For the **Demand Forecast** dashboard, I used the existing Total Demand measure to analyze historical passenger demand. Power BI's built-in forecasting feature was added to extend the historical demand trend and provide a projected continuation of the observed pattern.

Anomaly detection was also added to identify unusual observations in the demand trend. A **Demand Category** calculated column was created to classify passenger demand into **Low, Medium and High** categories based on passenger-count values.

The final page was created to bring together the important findings, observations, recommendations and overall conclusion of the project. The dashboard pages were also reviewed for consistent formatting, visual organization and interactive filtering.

9.3 Milestone 4 Dashboard Explanation

The Milestone 4 dashboard completes the project by extending the previous traffic, geographic and demand analysis into **executive intelligence, mobility equity, forecasting and final reporting**.

The **Executive Intelligence** dashboard begins with KPI cards showing **Total Demand, Total Trips, Traffic Reliability and Average Travel Time**. These cards provide a quick overview of the main mobility indicators before moving into detailed analysis.

The **Demand Trend** line chart shows how passenger demand changes over time. This helps users observe variations in demand across the available dates.

The **Trips by Transport Mode** chart compares **cars, bikes and buses** across routes. This provides a simple view of how different transport modes contribute to mobility activity across the network.

The **Reliability by Mode** visual presents the available reliability indicator across routes. Since the dataset does not contain separate incident information for each transport mode, the analysis uses the project-defined reliability measure rather than presenting unsupported mode-specific reliability values.

The **Travel Time by Area** chart compares average travel time across different locations. This helps users understand how travel conditions vary geographically.

The **Demand by Location** chart compares passenger demand across locations and provides a location-level view of mobility activity.

The **Mobility Equity** dashboard focuses on access and mobility conditions across different areas. The **Mobility Access Index** provides a project-defined score based on available demand, public transport usage and traffic reliability indicators.

The **Mobility Access by Area** chart compares the access index across locations. This allows users to examine differences in the calculated mobility-access indicator between areas.

The **Equity Matrix** combines the **Mobility Access Index, Traffic Reliability % and Public Transport Usage** for each location. This provides a combined view of access, reliability and public transport activity.

The **Accessibility Status** visual presents the distribution of the mobility access index across locations. It provides another way to examine the variation in calculated access levels.

The original requirement for **Mobility Access by Demographic Group** could not be implemented because the available dataset does not contain demographic fields. Similarly, Average Fare was not included because fare information was not available in the dataset.

The **Demand Forecast** dashboard focuses on historical and future demand patterns. The **Total Demand** KPI provides the overall passenger demand recorded in the dataset.

The **Historical Demand** line chart shows actual passenger demand across the available dates. This provides the foundation for the forecasting analysis.

The **Demand Forecast** feature uses Power BI's built-in forecasting capability to extend the historical demand trend and provide an estimated continuation of the observed pattern.

The **Anomaly Detection** feature identifies unusual observations in the demand trend. This helps highlight demand points that differ from the normal pattern identified by Power BI.

The **Demand Category** visual classifies passenger-count observations into **Low, Medium and High** categories. This provides a simple summary of how passenger demand is distributed across the available records.

The **Final Insights** page was designed as the closing section of the project. It brings together the main findings from the previous milestones, important observations, recommendations based on dashboard evidence and the overall project conclusion.

Overall, Milestone 4 completed the project by extending the original mobility dashboard into a broader **executive intelligence and analytical reporting solution**. The final project combines traffic analysis, geographic mobility, transport-mode analysis, weather-related insights, mobility access, demand forecasting and anomaly detection while maintaining transparency about the limitations of the available dataset.

9.4 Milestone 4 – Output Screenshots

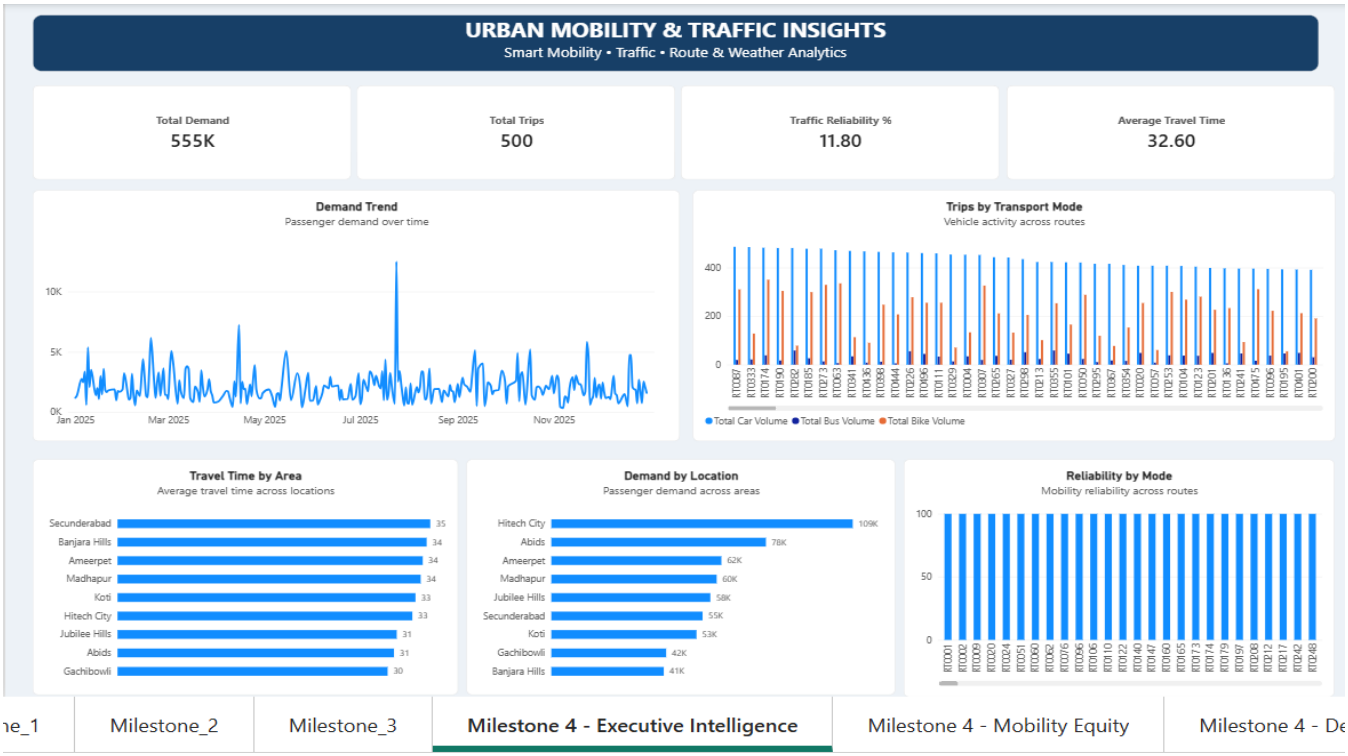


Figure 1. Milestone 4 – Executive Intelligence dashboard.

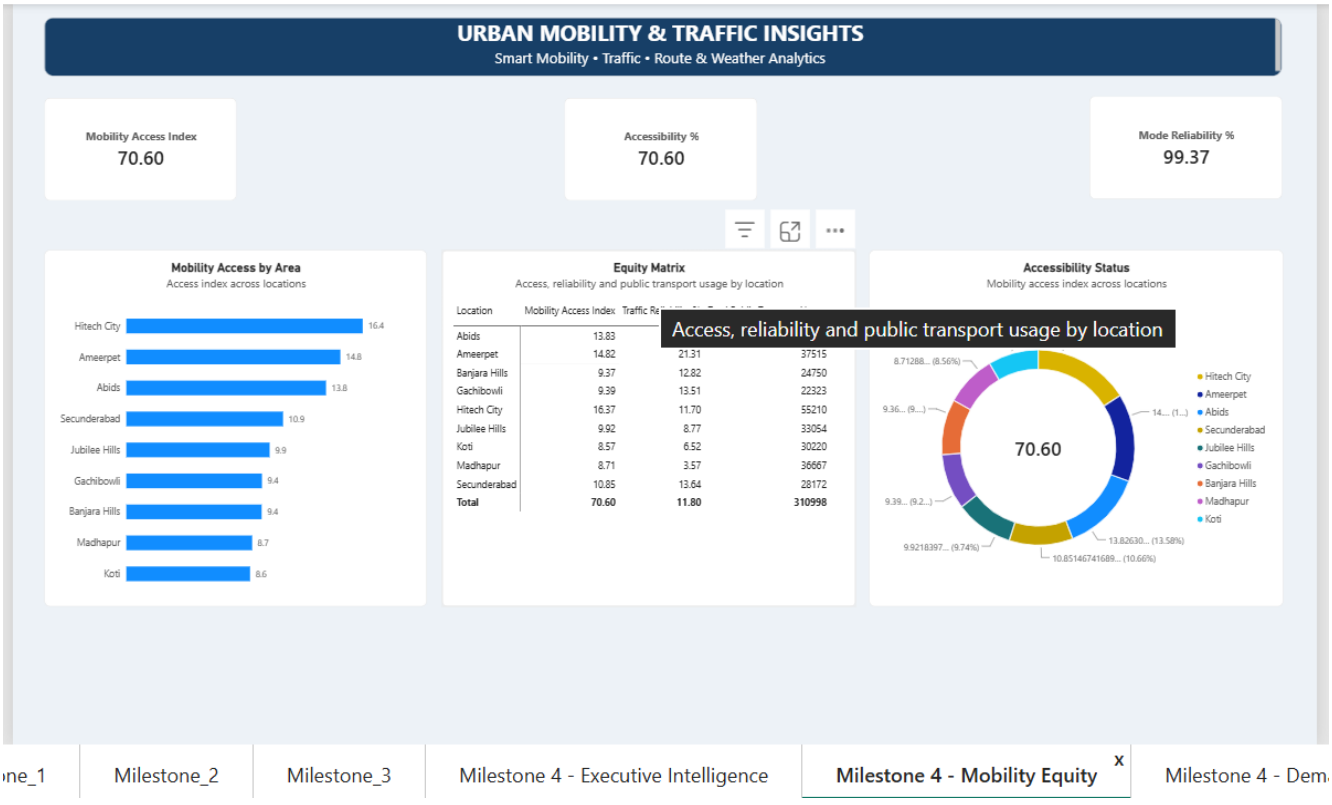


Figure 2. Milestone 4 – Mobility Equity dashboard.

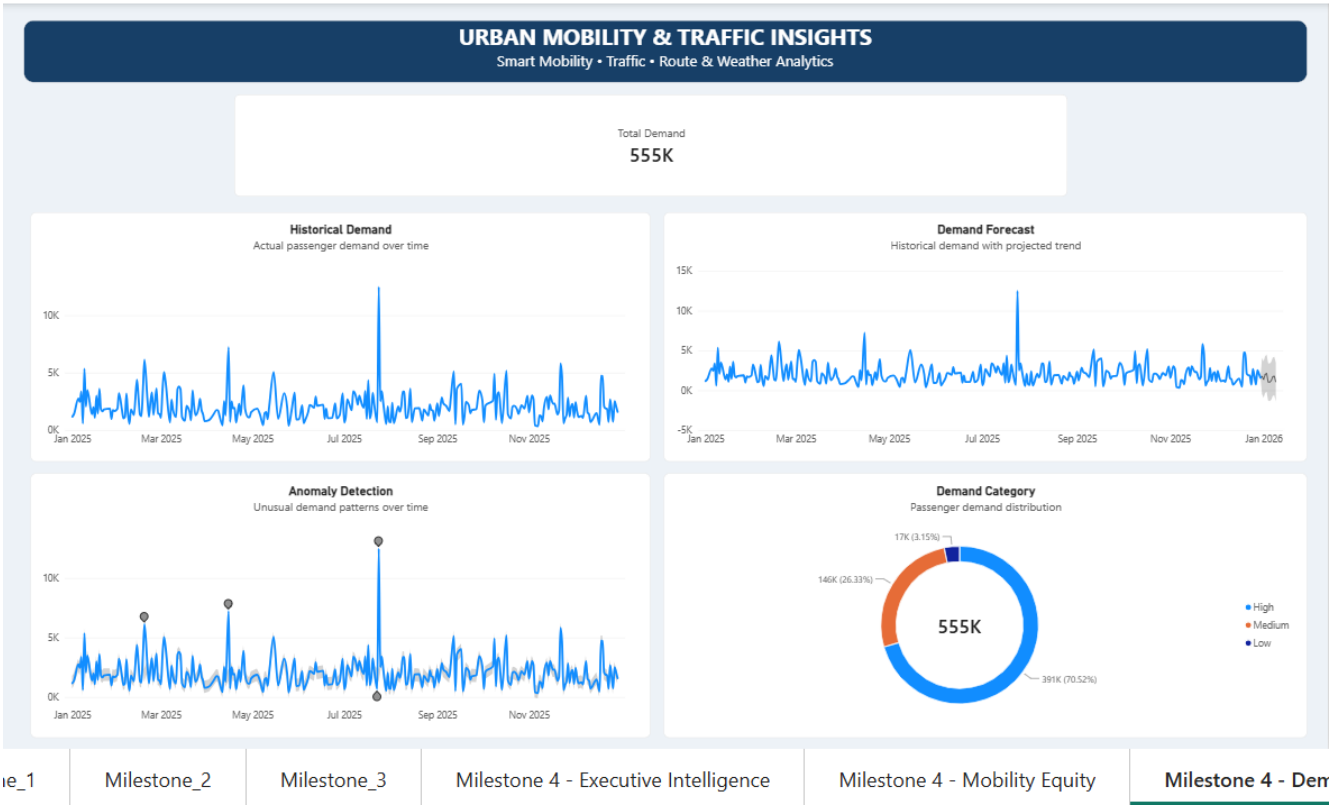


Figure 3. Milestone 4 – Demand Forecast dashboard.